

Lecture-2: Basic Asymptotics

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$= \frac{1}{2}n^2 + \frac{1}{2}n = O(n^2) = \frac{1}{2}n^2 + O(n)$$

↑
this means that it
grows no faster than
the function n

Recall: " $f(x) = O(g(x))$ (as $x \rightarrow \infty$)"

$$= \exists C, x_0 \text{ s.t. } |f(x)| \leq C \cdot g(x) \quad \forall x \geq x_0$$

(putting abs sign allows possibility of $f(x)$ being a negative function)

Usually $g(x)$ is a simpler function than $f(x)$, and eventually for large enough x , $f(x)$ is basically no bigger than $g(x)$ - up to a constant factor C .

Or you could write:

$$= \exists C, x_0 \text{ s.t. } 0 \leq f(x) \leq C \cdot g(x) \quad \forall x \geq x_0$$

We have a similar notation for when the parameter x is going to 0, usually as a positive number.
($x \rightarrow 0^+$)

$$= \exists C, x_0 > 0 \text{ s.t. } 0 \leq f(x) \leq C \cdot g(x) \quad \forall 0 < x \leq x_0$$

Usually in this scenario, we use a number that's denoting a small number like ϵ or δ .

The whole situation of definitions around Big O notation is a little bit slippery; but people have proposed the use of the following extension.

Extension: $O(g(x))$ denotes an anonymous function $f(x)$ s.t.
 $0 \leq f(x) \leq C \cdot g(x)$ for some $C, x \geq x_0$.

Another way to write

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} = \frac{1}{2}n^2 + \frac{1}{2}n = \frac{1}{2}n^2 + O(n)$$

$$= \frac{1}{2}n^2 \left(1 + O\left(\frac{1}{n}\right)\right) \rightarrow \Theta(n^2)$$

here, it's nice to have
an expression going to
1 as n gets large

In TCS, every time you have a complicated expression, you're trying to roughly understand how it grows or shrinks.

Recall : O is like $\leq$ Ω is like $\geq$ Θ is like $\approx$

analogy

opposites

DEFINITIONS

$$f(x) = \Omega(g(x)) : \exists c > 0, x_0 \text{ s.t. } f(x) \geq c \cdot g(x) \quad \forall x \geq x_0$$

$$f(x) = \Theta(g(x)) : f(x) = O(g(x)) \ \& \ f(x) = \Omega(g(x))$$

If you have a complicated expression $f(x)$, your first step should be determine a simple function $g(x)$, such that $f(x) = \Theta(g(x))$

Because then you've exactly nailed down the rate at which it grows, upto a constant factor.

Recall :

analogy

"little o"

o

is like

$<$

"little omega"

ω

is like

$>$

← opposites →

Eg. $f(x) = o(g(x))$

if $\frac{f(x)}{g(x)} \rightarrow 0$

e.g. $10n = o(n^2)$

← not only does $f(x)$ grow more slowly than $g(x)$ - it does so in a significant-enough way.

But $g(x)$ is asymptotically even greater than $f(x)$

More notation :

" $f(x) = \text{poly}(g(x))$ " $\equiv g(x)^{O(1)}$ (g of x to the big $O(1)$)

it's saying that $f(x)$ is AT MOST $g(x)$ to some constant power

→ O is something that denotes an anonymous function

that is eventually bounded by some fixed universal constant

Note: It's always better to state if x is going to $0/\infty$.

← used when you cannot exactly figure out what $f(x)$ is, but you've more or less figured it out

" $f(x) = \tilde{O}(g(x)) \equiv f(x) = g(x) \cdot \text{poly}(\log g(x))$ "

it's mostly like $O(g(x))$, but you allow yourself some logarithmic factors in the expression

Eg. $n^2 \cdot \log^3 n = \tilde{O}(n^2)$

↳ "it's more or less like quadratic running time"

$n^5 \cdot 3^n = \tilde{O}(3^n)$ ($\log 3^n = n \rightarrow$ so you're allowing yourself some extra powers of n here)

Q. $\stackrel{?}{=} \tilde{O}(2^n)$: 2^n grows much more slowly than 3^n . But you could look at the ratio between them, which is 1.5^n , and check that that's not as small as $\text{poly}(n)$.

(which it's not)

$\therefore$ There is a difference between 3^n and 2^n .

when you have a situation where $x \rightarrow 0$, $g(x) \rightarrow 0$

then $f(x) = \tilde{O}(g(x))$ means $f(x) = g(x) \text{ poly log } \left(\frac{1}{g(x)}\right)$

e.g. $f(\varepsilon) = \varepsilon^2 \cdot \log \frac{1}{\varepsilon}$
 $= \tilde{O}(\varepsilon^2)$

this function is basically like ε^2 , only a little bit bigger by a logarithmic factor.

$\uparrow$
this expression is sometimes called quasi-linear

e.g. $f(x) = \tilde{\Omega}(g(x))$ means $f(x) \geq \frac{g(x)}{\text{poly log}(g(x))}$

e.g. $\frac{n^3}{\log^3 n} = \tilde{\Omega}(n^3)$

The exact definition of big O notation when there's more than one variable is kind-of slippery.

$O_k(n^2)$ usually means there are 2 parameters k and n^2

and usually this means: order of n^2 , if you treat k as a complete constant

★ e.g. $2^{2^{(k^2)}} \cdot n^2 = O_k(n^2)$

Def: A function $g(n)$ is in standard form if it is a product of:

① constant (e.g. 6, $\sqrt{2\pi}$, ...)

note: $\ln n$
 $= \log_e n$

② constant powers of $\ln n$
(e.g. $\ln^2 n$, $\frac{1}{\sqrt{\ln n}}$, ...)

$\lg n = \log_2 n$

If you're writing a paper and you write $\log n$, and it makes a difference to you if the base is 2 or e , then you should SAY (on the paper).

↓

often, though, it doesn't come up, because if you're not trying to bound things very carefully, then all the logs are the same-

in the sense that $\ln n = \Theta(\log_2 n)$

continued...

③ constant powers of n (e.g. n^3 , $\frac{1}{n^2}$, ...)

④ exponential functions (e.g. 2^n , 3^n , $2^{n/2}$, ...)

⑤ $n^{c \cdot n}$, where c is a constant

↗
(this does not exhaust all kinds of functions).

But in many cases, when you're trying to bound something,

the nice bound ← that your function is big O over big Θ of

will be expressed as some product of these kinds of functions.

e.g. (in standard form):

$n^5 \cdot 3^n$, $6^{n^2} \sqrt{\log n}$, $\sqrt{2\pi} \sqrt{n} n^n e^{-n}$

↓
has all 5 kinds in it

good thing
to know →

↓
apparently Sterling's approximation
says that n factorial is very
close to this quantity

For these particular 5 examples above, each type is asymptotically smaller than the next type, even with arbitrary positive powers.

$$\text{So, } (\ln n)^{100} = o(n^{\frac{1}{10}})$$

↑ (type-2) is asymptotically smaller than (type-3)

$$\begin{matrix} \text{(type-3)} & & \text{(type-4)} \\ \hookleftarrow 100 n^{50} & = & o(1.1^n) \end{matrix}$$

Note: even within one type, if you have a higher positive power, you're asymptotically bigger.

$$\text{Eg. } 2^n = o(3^n)$$

$$(2^{(\log_3 2)^n}) ?$$

↑
little 'o' of means asymptotically smaller than